

Algrithm



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6. PORTFOLIO CONSTRUCTION & SYSTEM DEATH JULY 22, 2026 8 MIN PREMIUM

6.50 Crypto TSMOM Lives in Volume-Weighted Returns

Crypto time series momentum only pays when the market return is weighted by trading volume, not manipulable market cap: 0.94%/day, Sharpe 2.17. Equal-weight the same signal and it loses 1.19%/day.



A crypto momentum strategy that earns 0.94% a day at an annualized Sharpe of 2.17 should set off every alarm you own. Huang, Sangiorgi, and Urquhart report exactly that, buying coins that went up and shorting coins that went down, rebalanced daily across 3,192 cryptocurrencies from 2014 to 2023. Before you wire money, look at one number they also report: run the identical momentum signal with equal weights instead of volume weights and it loses 1.19% a day. The signal does not carry the strategy. The weighting scheme does. That single contrast is the whole paper, and it is both the interesting finding and the reason to be careful.

Time series momentum, in the old article "Building a Trend Follower, Component by Component," is the plainest edge in trading: assets that went up tend to keep going up, so you go long positive past returns and short negative ones. Moskowitz, Ooi, and Pedersen documented it across stocks, bonds, currencies, and commodities. In crypto the record was a mess, with some papers finding strong momentum and others finding none. This paper argues the disagreement comes

from a preprocessing choice nobody scrutinized: how you weight the coins when you build the market return.

Volume, not market cap, because market cap is a lie

The standard crypto "market return" weights each coin by market capitalization, following Liu and Tsyvinski. The authors refuse to, and the reason is specific. A coin's market cap is supply times price, and supply is whatever the founders say it is. Mint more coins, burn coins to manufacture scarcity, and the cap moves without a single trade happening. They point to coins like Peercoin and Namecoin carrying market caps above 1 million dollars while trading under 10,000 dollars a day. Cap says "large," the tape says "nobody is here." Volume is harder to fake with a keystroke, so they weight by trading volume instead.

$$R_{VW,t} = \sum_{i=1}^N \frac{v_{i,t}}{\sum_{j=1}^N v_{j,t}} r_{i,t}$$

Read it as a weighted average: each coin i gets a weight equal to its dollar volume v that day divided by the total volume of every coin, then you sum those weighted returns. Worked example with three coins on a given day. Bitcoin trades 40 billion and returns +2%, Ethereum trades 15 billion and returns +1%, some alt trades 1 billion and returns +30%. Total volume is 56 billion. The volume-weighted return is $(40/56)(2\%)$ plus $(15/56)(1\%)$ plus $(1/56)(30\%)$, which is 1.43% plus 0.27% plus 0.54%, so about 2.24%. The alt's 30% pop barely registers because almost nobody traded it. Under equal weighting that same alt would contribute a full 10%, letting a coin nobody touched swing the index.

The choice matters before any strategy runs. The volume-weighted market averages 0.62% a day over the sample, against 0.21% for cap-weighted, 0.33% for equal-weighted, and 0.11% for Bitcoin alone. The index you pick already decides most of your answer.

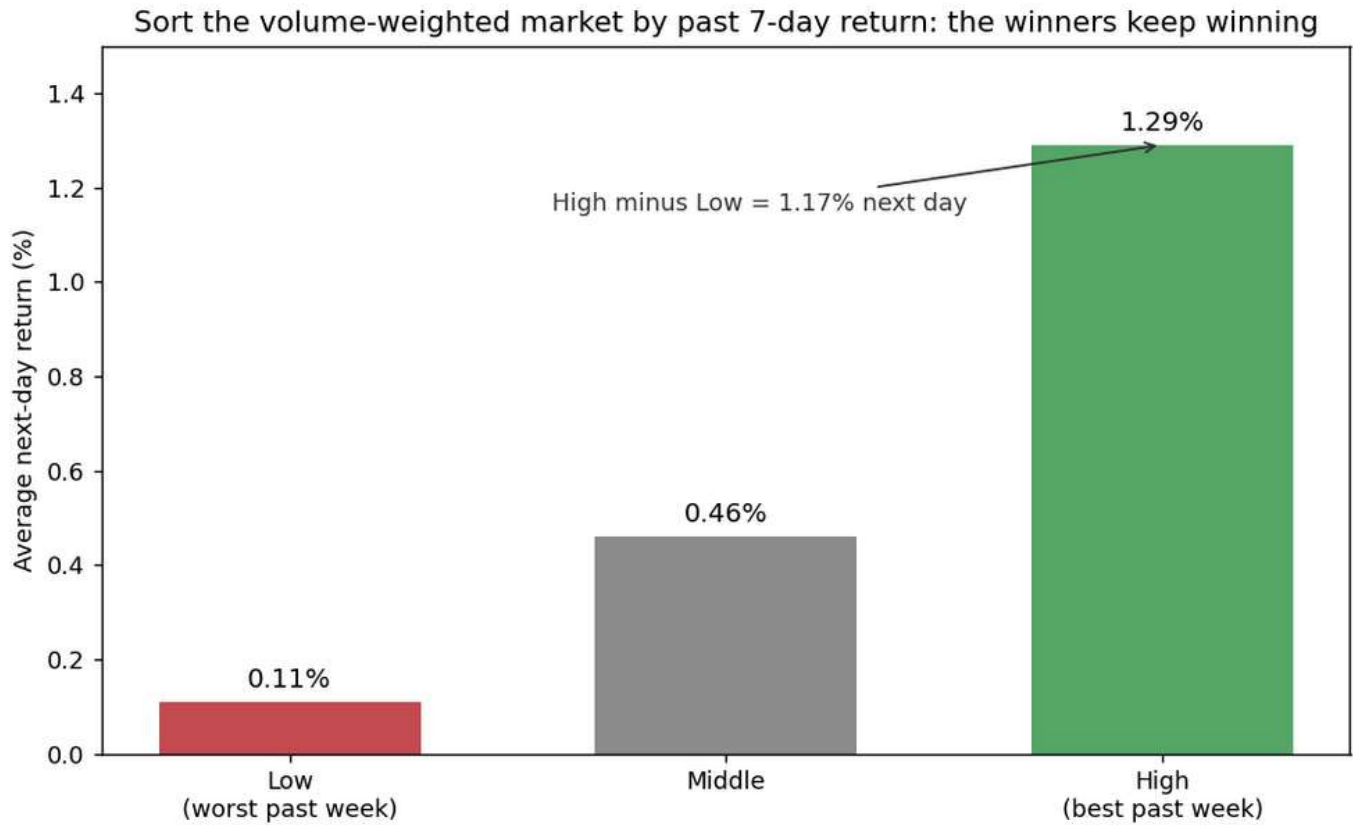
The momentum is real, at least inside the index

They test time series momentum first by regressing the next day's volume-weighted market return on its own past cumulative return over horizons from one day to two weeks.

$$R_{VW,t} = \alpha + b R_{VW,t-1:t-h} + e_t$$

The slope b answers whether the past h -day return predicts tomorrow. Positive and significant means momentum; the intercept α is the part unrelated to the past, and e is noise. The slope lands between 0.024 and 0.042 and is significant at the 1% level from the 3-day horizon out to two weeks, across the full sample and both subperiods. Worked reading: a b of 0.04 means that for every 1% of cumulative return over the past week, tomorrow's expected return rises by 0.04%. Small per unit, but past weekly moves are often tens of percent in crypto, so it adds up. The one-day and two-day horizons are insignificant, which fits the story that a single day is mostly noise.

Sorting sharpens it. Group each day by the tercile of past return and look at the next day. The bottom tercile earns almost nothing the next day, the middle earns more, and the top tercile earns the most, a clean monotonic ladder that holds even when the tercile cutoffs are fixed using only 2014 to 2015 data and applied out of sample through 2023.



The High-minus-Low gap runs above 1% a day and the weekly version peaks near 2.9% in-sample and above 4% out-of-sample. Continuation is there.

Building the long-short, and where the number comes from

The tradable version is a Winner Minus Loser portfolio. Split coins into winners (positive past-formation return) and losers (negative), weight each side by volume, go long winners and short losers, hold one day, rebalance.

$$w_{i,t} = \frac{v_{i,t}}{\sum_{k \in \text{side}} v_{k,t}}, \quad R_{\text{WML},t} = \sum_{i \in W} w_{i,t} r_{i,t} - \sum_{j \in L} w_{j,t} r_{j,t}$$

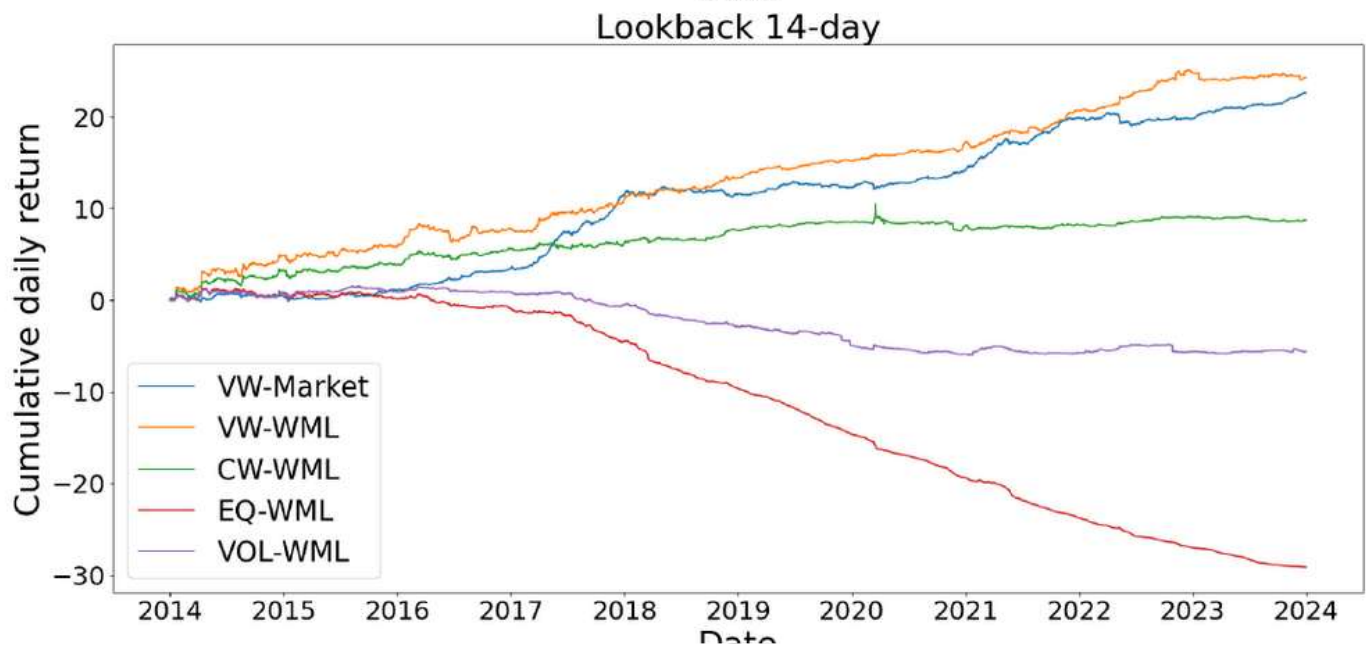
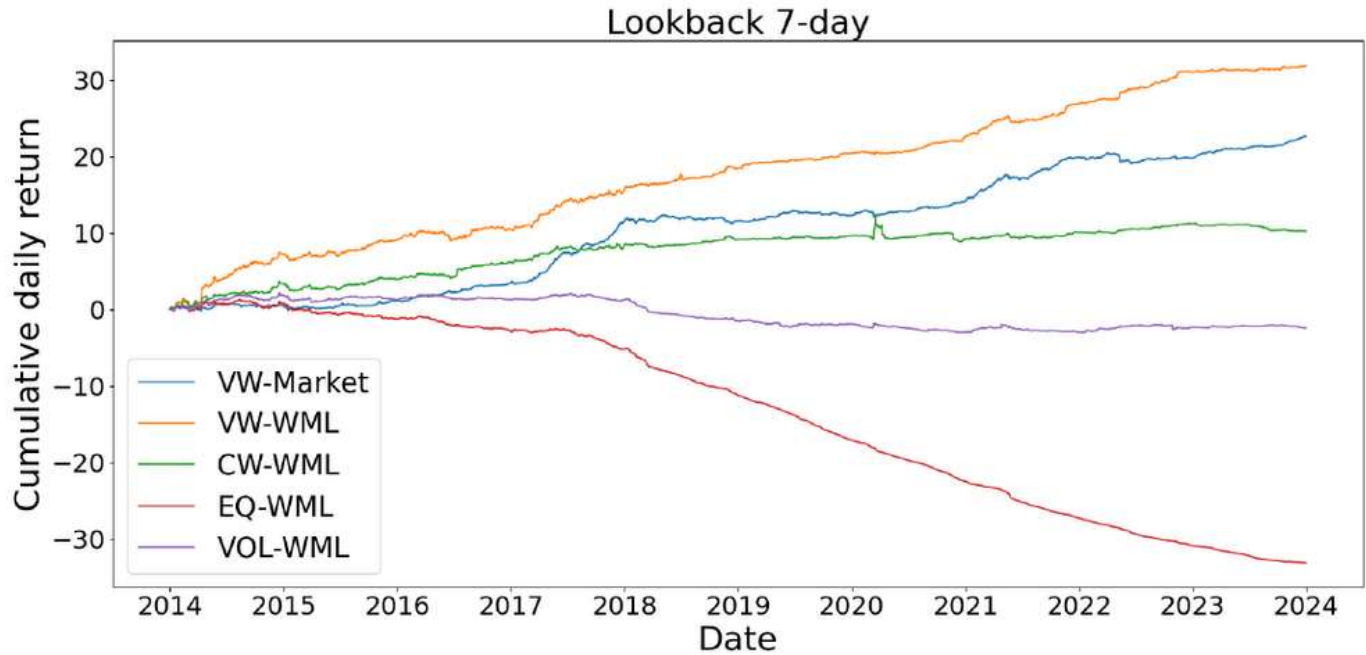
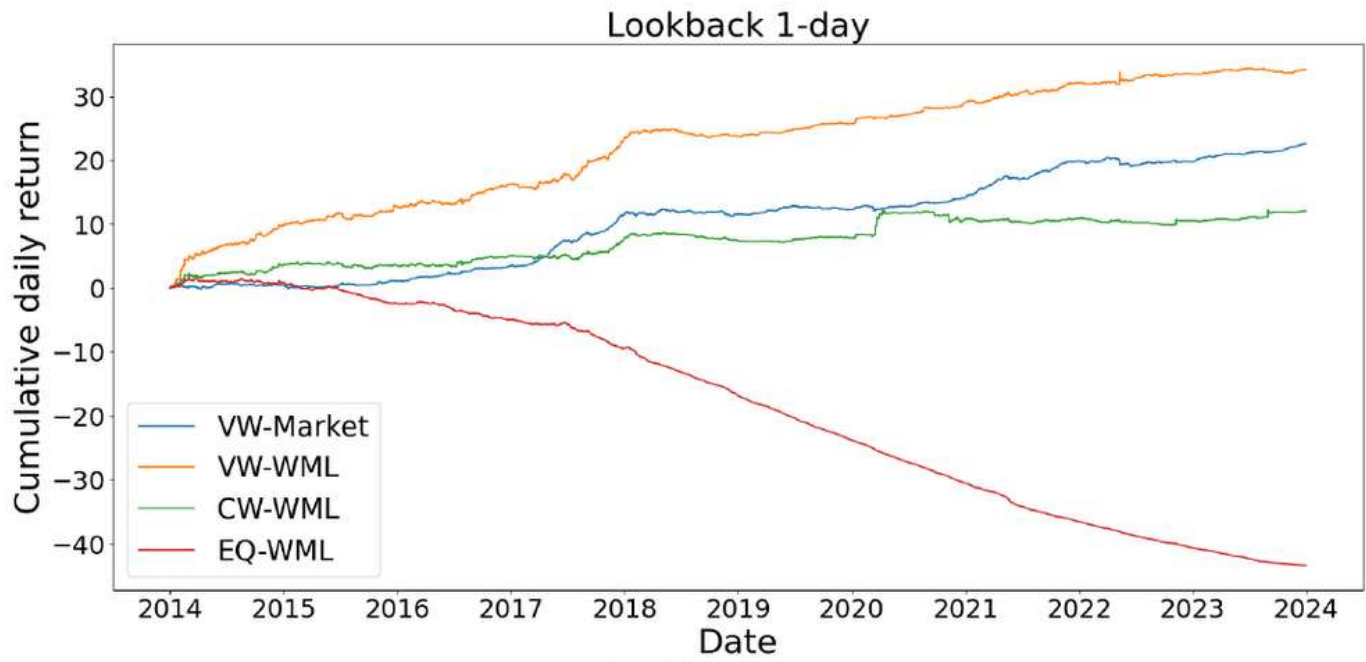
Inside each leg a coin's weight is its volume over the total volume of that leg; the long-short return is the volume-weighted winner return minus the volume-weighted loser return. Worked example: the winner leg is 70% Bitcoin and 30% Ethereum by volume and returns 0.70(1.5%) plus 0.30(1.0%) equal to 1.35%; the

loser leg similarly weighted returns 0.4%; the WML earns 1.35% minus 0.4%, so 0.95% that day. That is the mechanism behind the headline 0.94% (1-day formation), 0.87% (7-day), and 0.67% (14-day).

Risk-adjust it with the Sharpe ratio, annualized across crypto's 365 trading days.

$$SR_p = \frac{\overline{R}_p - R_f}{\sigma_p}$$

Mean excess return over standard deviation. Worked reading: an annualized Sharpe of 2.17 on the 1-day portfolio means the strategy earned 2.17 units of return per unit of volatility per year, which for a daily-rebalanced crypto book is very high. The 7-day formation posts the best Sharpe at 2.66, the 14-day drops to 1.86.



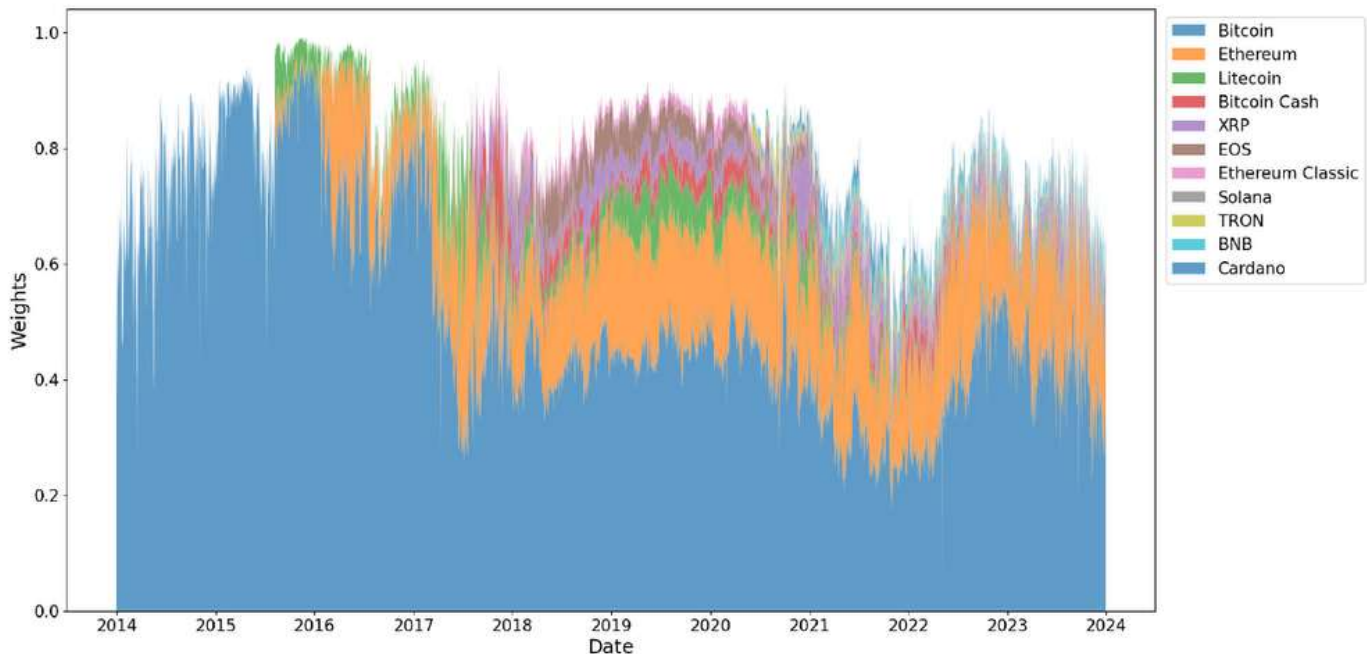
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The volume-weighted WML (orange) compounds above everything else, more than doubling the cap-weighted WML (green). Hold that picture, because the two lines heading down are the point.

The weighting scheme is the strategy

Equal-weighted WML loses 1.19% a day. Inverse-volatility WML, the sizing rule the old article "Building a Trend Follower, Component by Component" recommends for classic trend systems, also loses money here. Only volume weighting and, to a weaker extent, cap weighting make the long-short profitable. That is not a footnote. It says momentum among the typical coin, the small illiquid one that equal weighting surfaces, does not continue. Continuation lives in the coins that trade the most, and volume weighting is a device for concentrating the book into them.

And what trades the most is Bitcoin. Across the sample Bitcoin is 52.62% of total volume and Ethereum another 16.82%.



So the honest description is narrower than "crypto has a volume-weighted momentum factor." It is closer to "a book dominated by Bitcoin and Ethereum

momentum, financed by shorting small coins, makes money, while momentum spread evenly across the long tail does not." The volume weight is doing the same job market cap does in equities, tilting hard toward a few giants, except the giants here also happen to be the ones where a daily trend is real and tradable. This connects to the old article "Order Flow Prices the Crypto Cross-Section (Permanently)": genuine trading activity, not stated supply, is what carries the signal, and volume is the cleaner proxy for it.

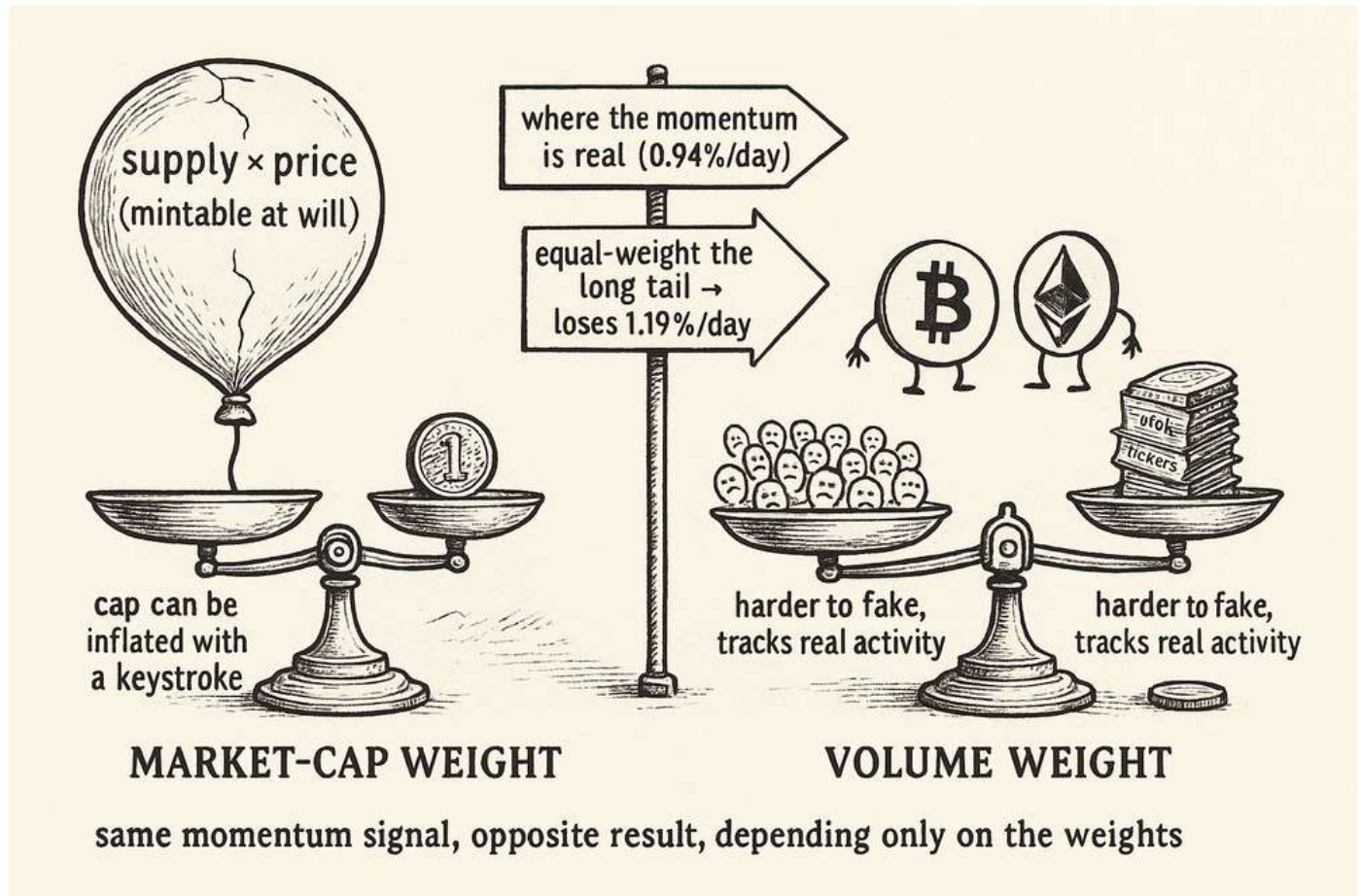
Costs, shorts, and wash trades

The authors are not naive about costs. They rerun with a fixed 10 basis point transaction cost and the volume-weighted WML still clears 0.74% (1-day), 0.67% (7-day), and 0.47% (14-day), with factor alphas against the Liu-Tsyvinski-Wu market, size, and momentum factors staying significant at 0.4% to 2%. The strategy is not subsumed by cross-sectional momentum or MAX momentum either. On paper it survives.

The gaps between paper and P&L are wide, and they all sit on the short side. A daily-rebalanced long-short over 3,192 coins turns the whole book every day; 10 basis points round-trip is generous for large caps and fantasy for the small alts in the loser leg, where spreads are wide and market impact is real. Worse, you cannot short most of those coins at all, because there is no borrow, so the loser leg that contributes a chunk of the spread is often not investable. And the volume weights themselves lean on data the authors admit is dirty: crypto exchanges wash-trade, inflating exactly the volume figures that set the portfolio weights. They check this by rebuilding with Coin Metrics "trusted exchange" volume and the result holds qualitatively, which is reassuring but not the same as clean. Treat the 0.94% as a gross, long-friendly, upper bound, not a number you will see in an account.

None of that kills the finding. It reframes it. The contribution is not a printing press; it is a diagnostic. If you are going to compute a crypto market return or a crypto momentum signal, weight by volume rather than by market cap, because

cap is manipulable and volume tracks where trading actually happens. The tradable edge that remains after real costs and real short constraints is far smaller than the headline, and it is concentrated in the handful of coins liquid enough to matter.



KEY POINTS

- Crypto time series momentum works when you build the market return by trading volume, not market cap. Volume-weighted WML earns 0.94% a day (1-day formation) at an annualized Sharpe of 2.17, and 0.87% at Sharpe 2.66 for the 7-day formation.
- Market cap is manipulable because supply is discretionary; coins carry million-dollar caps on under 10,000 dollars of daily volume. Volume is the cleaner proxy for real activity. The volume-weighted market averages 0.62% a day versus 0.21% cap-weighted.

- Momentum in the index is genuine: the next-day regression slope is 0.024 to 0.042 and significant from 3 days to 2 weeks, and past-return terciles produce a monotonic next-day ladder that holds out of sample.
- The weighting scheme is the strategy, not the signal. Equal-weighted WML loses 1.19% a day and inverse-volatility WML also loses. Continuation lives in high-volume coins, and Bitcoin plus Ethereum are about 70% of volume.
- Alphas against the market, size, and momentum factors stay at 0.4% to 2% after a 10 basis point cost, and the effect is not subsumed by cross-sectional or MAX momentum.
- Take the headline as a gross upper bound. Daily rebalancing over 3,192 coins with a short leg of tiny alts faces costs far above 10 basis points, frequent no-borrow, and wash-trading-inflated volume weights. The trusted-exchange robustness check helps but does not make the data clean.

A note on AI. *The ideas, research, analysis, and conclusions in this article are my own. I use AI tools to help with editing and wordsmithing, because English is not my first language, and I am not shy about that. AI-generated ideas and AI-assisted writing are not the same thing: the first is empty slop from a generic prompt, the second is a tool for communicating years of real research more clearly. Judge the work by its substance, not by whether software helped polish the prose.*

References

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